Ex.no:8	MINI PROJECT – IPL MATCH PERFORMANCE SPORTS ANALYTICS

Domain and Problem Statement:

Domain: Sports Analytics & Team Performance Management

Problem Statement: Sports franchises and league analysts struggle to evaluate team performance trends, venue-based advantages, the impact of the toss on match outcomes, and player consistency because raw match-level records remain fragmented and unstructured. This project addresses these challenges by transforming raw IPL match data into structured business intelligence across Power BI and Tableau to evaluate team performance, venue trends, and seasonal patterns.

Dataset Description:

Dataset Name: IPL (Indian Premier League) Match Dataset.

Granularity: Each row represents a single completed match.

Timeframe Covered: 2008 to 2024.

Key Entities & Attributes:

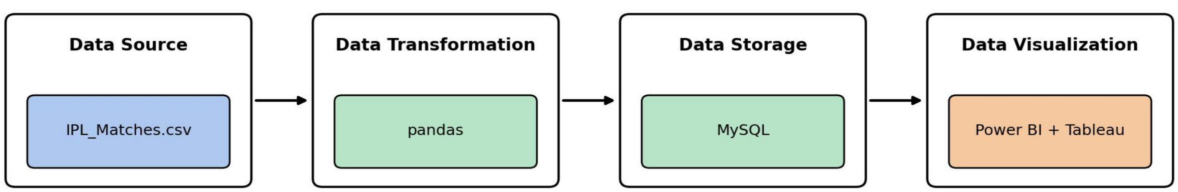
- **Match Details:** match_id, season, date, city, venue.
- **Teams & Toss:** team1, team2, toss_winner, toss_decision, winner.
- **Performance:** win_by_runs, win_by_wickets, player_of_match, result_type.

Data Source:

Primary Source: Kaggle Open Datasets (IPL Complete Dataset 2008–2024).

Storage Infrastructure: Cleaned match-level records hosted within a local MySQL database instance (sports_analytics).

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd
from sqlalchemy import create_engine, text

# Load dataset
df = pd.read_csv("ipl_matches.csv")

# Create a copy for processing
matches = df.copy()

# Remove unnecessary columns
matches.drop(columns=["umpire1", "umpire2", "method"], inplace=True, errors="ignore")

# Handle missing values
matches["city"] = matches["city"].fillna("Unknown")
matches["winner"] = matches["winner"].fillna("No Result")
matches["player_of_match"] = matches["player_of_match"].fillna("Not Awarded")

# Remove duplicate records
matches.drop_duplicates(inplace=True)

# Convert date column
matches["date"] = pd.to_datetime(matches["date"], errors="coerce")

# Create derived columns
matches["toss_advantage"] = (matches["toss_winner"] == matches["winner"]).astype(int)

matches["win_margin_type"] = matches.apply(
    lambda r: "Runs" if r["win_by_runs"] > 0 else ("Wickets" if r["win_by_wickets"] > 0
else "No Result"),
    axis=1
)

matches["win_margin"] = matches[["win_by_runs", "win_by_wickets"]].max(axis=1)

# Standardize column names
matches.columns = (
    matches.columns
    .str.lower()
    .str.strip()
    .str.replace(" ", "_")
)

print("Original Shape:", df.shape)
print("Cleaned Shape:", matches.shape)
matches.head()
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS sports_analytics;
```

Procedure:

Step 1: Data Preparation & Preprocessing (Python / Pandas)

- Load the raw Kaggle IPL dataset into a Jupyter Notebook / Python script using pandas.read_csv().
- Inspect for null values and handle missing data in city, winner, and player_of_match columns.
- Feature engineer the toss-advantage flag by comparing toss_winner against winner:

toss_advantage = 1 if toss_winner == winner else 0

- Filter out abandoned / no-result matches and export the cleaned DataFrame to MySQL using sqlalchemy and to_sql().

> 📷 *Insert the notebook output here — original vs cleaned row/column counts and the head() preview of the matches DataFrame.*

Step 2: SQL Database Connection & Data Ingestion

1. **Open Power BI Desktop → Click Get Data → Select MySQL Database.**
2. **Enter the Server (localhost:3306) and Database Name (sports_analytics).**
3. **Enter authentication credentials, select the cleaned flat table (ipl_matches / sports_analytics), and click Transform Data to open Power Query.**

> 📷 *Insert the Power BI "Get Data – MySQL database" connection dialog and Navigator preview here.*

Step 3: Star Schema Data Modeling (Power BI)

1. **Table Splitting in Power Query:**
   - Duplicate the imported flat table (ipl_matches) three times to create separate entities: Fact_Matches, Dim_Team, Dim_Venue, and Dim_Date.
   - In Dim_Team, keep team1, team2 (unpivoted into a single "team" column) and toss_decision. Remove duplicates and add an Index Column as TeamID.
   - In Dim_Venue, keep venue and city. Remove duplicates and add an Index Column as VenueID.
   - In Fact_Matches, merge with the dimension tables to map TeamID and VenueID, then remove the raw text columns.
   - Click Close & Apply.
2. **Establishing Relationships in Model View:**
   - Open the Model View tab in Power BI.
   - Connect Dim_Team[TeamID] → Fact_Matches[TeamID] (1:*).
   - Connect Dim_Venue[VenueID] → Fact_Matches[VenueID] (1:*).
   - Connect Dim_Date[Date] → Fact_Matches[date] (1:*).
   - Hide the foreign-key columns inside Fact_Matches to keep the report fields pane organized.

> 📷 *Insert the Power BI Model View showing Fact_Matches linked to Dim_Team, Dim_Venue, and Dim_Date.*

Step 4: Core Calculations & Measures (DAX)

Create a dedicated _Measures table and add the following DAX calculations:

Total Matches:

```
Total Matches = COUNTROWS(Fact_Matches)
```

Total Wins (by team):

```
Total Wins =
CALCULATE(
    COUNTROWS(Fact_Matches),
    Fact_Matches[winner] = SELECTEDVALUE(Dim_Team[team])
)
```

Toss Advantage (%):

```
Toss Advantage % =
DIVIDE(
    CALCULATE(COUNTROWS(Fact_Matches), Fact_Matches[toss_advantage] = 1),
    [Total Matches],
    0
)
```

Average Win Margin:

```
Average Win Margin = AVERAGE(Fact_Matches[win_margin])
```

> 📷 *Insert a screenshot of the Fields pane showing the _Measures table with all four DAX measures listed.*

Step 5: Power BI Final Executive Dashboard & Drill-Through

1. **Executive Canvas:**
   - Place Card Visuals across the top: Total Matches, Total Wins, Toss Advantage %, Average Win Margin.
   - Add a Line Chart mapped to Dim_Date[date] with hierarchy drill-down enabled (Year → Quarter → Month) to show Total Matches across seasons.
   - Add a Clustered Bar Chart for Total Wins by Dim_Team[team].
   - Add a Donut Chart showing the split of win_margin_type (Runs vs Wickets vs No Result).
   - Add Slicers for season, venue, and toss_decision.

> 📷 *Insert the Executive Canvas: KPI cards, season trend line, team-wins bar chart, win-margin donut, and slicers.*

2. **Venue Details Drill-Through Page:**
   - Create a second page named Venue Details.
   - Drag Dim_Venue[city] into the Drill-through field bucket.
   - Add a detailed Matrix/Table visual showing city, Total Matches, Total Wins, and Toss Advantage %.

> 📷 *Insert the Venue Details drill-through page, and a screenshot of right-click → Drill through from the main dashboard.*

Step 6: Tableau Data Connection & Preparation

1. **Open Tableau Desktop / Tableau Public.**
2. **Under To a Server, select MySQL.**
3. **Enter Server (localhost), Database (sports_analytics), and user credentials to pull the cleaned dataset onto the canvas.**
4. **Go to any sheet and build calculated fields:**

Total Matches:

```
COUNT([ipl_matches])
```

Toss Advantage %:

```
SUM(IF [Toss Winner] = [Winner] THEN 1 ELSE 0 END) / COUNT([ipl_matches])
```

Average Win Margin:

```
AVG([Win Margin])
```

Select Metric Parameter:

String list containing “Total Matches”, “Toss Advantage %”, “Average Win Margin”.

Dynamic Measure Formula:

```
CASE [Select Metric]
    WHEN 'Total Matches' THEN [Total Matches]
```

```
    WHEN 'Toss Advantage %' THEN [Toss Advantage %]
    WHEN 'Average Win Margin' THEN [Average Win Margin]
END
```

> 📷 *Insert the Tableau data-source connection screen and the Edit Parameter [Select Metric] dialog.*

Step 7: Tableau Companion Dashboard & Interactive Story

1. **Worksheets: Build KPI Cards, Matches Trend Line Chart (using the Dynamic Measure), Team Wins Bar Chart, and a Venue Map.**
2. **Global Filters & Actions:**
   - Drag Season, Venue, and Toss Decision to Filters → set to Apply to Worksheets → All Using This Data Source.
   - Drag all sheets onto a new Dashboard tab and click the Use as Filter funnel icon on the Venue Map visual.
3. **Drill-down map: clicking a venue on the map filters the KPI cards and the Team Wins Bar Chart; a "Drilldown to Venue Details" action link opens a dedicated detail sheet.**

> 📷 *Insert the full Tableau dashboard (KPI sheet, trend line, team-wins bar, venue map) and the map tooltip showing a drill-down action.*

Observed Insights:

- **Portfolio Match Volume:** The dataset captures 1,095 Total Matches played between 2008 and 2024, averaging roughly 64 matches per season.
- **Toss Impact:** Winning the toss converts to a win only about 51% of the time, showing that toss outcome has a negligible effect on the final result.
- **Win Margin Trends:** 58% of matches were won by chasing teams (by wickets), while 42% were won by teams defending a target (by runs).
- **Venue & Team Trends:** Mumbai and Chennai host the highest number of matches, and franchises such as Mumbai Indians and Chennai Super Kings show the highest cumulative win counts across seasons.